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Air conditioning apparatus for vehicle

Abstract

The present invention relates to an air conditioning apparatus for a vehicle and, more specifically, to an air conditioning apparatus for a heat pump system, wherein while blown air passes through a first area and a second area which are partitioned from each other by a first separation wall, air passing through the first area is cooled or heated and then supplied to the inside of a vehicle, and air passing through the second area is discharged to the outside, so that the air conditioning apparatus can be miniaturized and can easily cool or heat the inside of the vehicle.

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Background/Summary

(1) This application is a national phase under 35 U.S.C. § 371 of International Application No. PCT/KR2019/010561 filed Aug. 20, 2019, which claims the benefit of priority from Korean Patent Application Nos. 10-2018-0103537 filed on Aug. 31, 2018, 10-2018-0103684 filed on Aug. 31, 2018 and 10-2018-0103821 filed on Aug. 31, 2018. The entire contents of these applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The disclosure relates to an air conditioning apparatus for a vehicle and, more particularly, to an air conditioning apparatus for a vehicle, in which while blown air passes through a first area and a second area which are partitioned from each other by a first separation wall, air passing through the first area is cooled or heated and then supplied to the inside of a vehicle, and air passing through the second area is discharged to the outside, and the air conditioning apparatus may thus be miniaturized and may easily cool or heat the inside of the vehicle.

BACKGROUND ART

(3) A general air conditioner system for a vehicle may include a compressor compressing and delivering a refrigerant, a condenser condensing the high-pressure refrigerant delivered from the compressor, an expansion member throttling the liquefied refrigerant condensed in the condenser, and an evaporator evaporating the low-pressure liquid refrigerant throttled by the expansion member through heat-exchange with air blown to the inside of the vehicle, thereby cooling air discharged to the inside using heat absorption by evaporative latent heat of the refrigerant, which are connected to one another through a refrigerant pipe.

(4) The evaporator may be installed in an air conditioning case installed on the inside the vehicle to serve a cooling function. That is, air blown by a blower may pass through the evaporator and may be cooled by the evaporative latent heat of the liquid refrigerant circulated in the evaporator and discharged to the inside of the vehicle in a cold state.

(5) In addition, heating the inside of the vehicle may use a heater core installed in the air conditioning case and circulating engine coolant, or an electric heating type heater installed in the air conditioning case.

(6) Meanwhile, the condenser may be installed on the front side of the vehicle to dissipate heat while exchanging the heat with air.

(7) In recent years, a heat pump system performing cooling and heating using only a refrigeration cycle has been developed, in which a cool air passage **11** and a hot air passage **12** are formed to be

partitioned in one air conditioning case **10**, an evaporator **4** cooling the air is installed in the cool air passage **11**, and a condenser **2** heating the air is installed in the hot air passage **12** as shown in FIG. **1**. Here, an air discharge port **15** supplying air to the inside of the vehicle and an air outlet port **16** discharging air to the outside of the vehicle may be formed at the outlet side of the air conditioning case **10**. In addition, blowers **20** operated individually from each other may be respectively installed on the inlet side of the cool air passage **11** and the hot air passage **12**.

(8) In a cooling mode, cool air cooled while passing through the evaporator **4** of the cool air passage **11** may thus be discharged to the inside of the vehicle through the air discharge port **15** to cool the inside of the vehicle. Here, hot air heated while passing through the condenser **2** of the hot air passage **12** may be discharged to the outside of the vehicle through the air outlet port **16**.

(9) In a heating mode, the hot air heated while passing through the condenser **2** of the hot air passage **12** may be discharged to the inside of the vehicle through the air discharge port **15** to heat the inside of the vehicle. Here, the cool air cooled while passing through the evaporator **4** of the cool air passage **11** may be discharged to the outside of the vehicle through the air outlet port **16**.

(10) However, the prior art has an increased overall size inevitably because each air outlet port **16** having an adjusting door is required to be formed in two positions as the blowers **20** are respectively positioned in the cool air passage **11** and the hot air passage **12**, and one of the air passing through the condenser **2** and the evaporator **4** needs to be discharged through the air outlet port.

(11) In addition, the condenser **2** installed in the air conditioning case is required to have a small size, and may thus have a low condensing performance and a problematic cooling performance.

PRIOR ART DOCUMENT

Patent Document

(12) Korean Patent Publication No. 10-1251206 (Mar. 29, 2013)

DISCLOSURE

Technical Problem

(13) An object of the present disclosure is to provide an air conditioning apparatus for a vehicle, in which while blown air passes through a first area and a second area which are partitioned from each other by a first separation wall, air passing through the first area is cooled or heated and then supplied to the inside of a vehicle, and air passing through the second area is discharged to the outside, and the air conditioning apparatus may thus be miniaturized and may easily cool or heat the inside of the vehicle.

(14) In more detail, an object of the present disclosure is to provide an air conditioning apparatus for a vehicle, in which: air is simultaneously blown to the first area and the second area by one blower; the first area communicates with the inside of the vehicle, and an evaporator and an internal heat exchanger are installed on the inside to cool and heat the inside; and the second area communicates with the outside, the heat exchanger is installed on the inside, and discarded air passing through the heat exchanger by cooling and heating settings is discharged to the outside without a separate door, thereby having a small size.

(15) In addition, another object of the present disclosure is to provide an air conditioning apparatus for a vehicle, which is installed in the mounting hole of a dash panel, is more easily mounted by having the evaporator, the heat exchanger and the blower installed in an engine room side in the air conditioning case and by having the internal heat exchanger disposed on the inside of the vehicle, and effectively utilizes a space in the engine room.

(16) In addition, another object of the present disclosure is to provide an air conditioning apparatus for a vehicle, in which outside air or inside air is selectively supplied to the first area through a first adjusting door of the air conditioning case, and the outside air is supplied to the second area through a second adjusting door.

(17) Here, another object of the present disclosure is to provide an air conditioning apparatus for a vehicle, in which an engine room air inlet is formed to be opened and closed so that air in the

engine room is introduced through the second adjusting door, and during maximum heating, air in the engine room, which has a higher temperature than the outside, is used to increase an evaporation performance (heat absorption amount) of the heat exchanger, thereby further enhancing a heating performance of the internal heat exchanger.

(18) In addition, another object of the present disclosure is to provide an air conditioning apparatus for a vehicle, in which an auxiliary inside-air inlet hole additionally supplying the inside air to the second area and a third adjusting door adjusting its flow are formed, and during the maximum heating, the inside warm air is used to further enhance the evaporation performance of the heat exchanger.

(19) In addition, another object of the present disclosure is to provide an air conditioning apparatus for a vehicle, in which an auxiliary heating heat exchanger is additionally installed behind the internal heat exchanger, thereby securing a faster and more effective heating performance.

Technical Solution

(20) In one general aspect, an air conditioning apparatus for a vehicle, includes: an air conditioning case **110** including a first area **A1** which communicates with the inside of the vehicle and in which inside-air conditioning wind flows and a second area **A2** which communicates with the outside of the vehicle; and a blower **160** which blows the wind to the first area **A1** and the second area **A2**.

(21) In addition, the blower **160** may be a blower operated by one motor, and the wind may be blown to the first area **A1** and the second area **A2** by the blower.

(22) In addition, the air conditioning apparatus **1000** for a vehicle may include an evaporator **130** for a cooling and an internal heat exchanger **120** for a heating in the first area **A1** of the air conditioning case **110**.

(23) In addition, the air conditioning apparatus **1000** for a vehicle may include a heat exchanger **140** installed in the second area **A2** of the air conditioning case **110** to condense a refrigerant during the cooling and to evaporate the refrigerant during the heating.

(24) In addition, when wind passing through the first area **A1** is cold wind for the cooling, wind passing through the second area **A2** may be wind having a higher temperature than the cold wind.

(25) In addition, when the wind passing through the first area **A1** is warm wind for the heating, the wind passing through the second area **A2** may be wind having a lower temperature than the warm wind.

(26) In addition, the air conditioning case **110** may include: a first outside-air inlet **114a** into which outside air is introduced and an inside-air inlet **116** into which inside air is introduced, which is each formed by making a certain area of the air conditioning case **110** hollow; a first adjusting door **191** adjusting openings and closings of the first outside-air inlet **114a** and the inside-air inlet **116**; a second outside-air inlet **114b** which is formed in the second area **A2** of the air conditioning case **110** and into which the outside air is introduced; and a second adjusting door **192** adjusting opening and closing of the second outside-air inlet **114b**.

(27) In addition, the air conditioning case **110** may include an engine room air inlet **115** which is formed by making a certain area hollow to allow the air in the engine room to be introduced into the second area **A2**, and opening and closing of which is adjusted by the second adjusting door **192**.

(28) In addition, the air conditioning case **110** may include an auxiliary inside-air inlet hole **118** which is formed by making a certain area hollow to allow the inside air to be introduced into the second area **A2**, and opening and closing of which is adjusted by a third adjusting door **193**.

(29) In addition, the auxiliary inside-air inlet hole **118** may be formed by making a certain area of a partition wall **110w** adjacent to the inside-air inlet **116** hollow, and some of the air introduced into the first area **A1** through the inside-air inlet **116** may be moved to the second area **A2** through the auxiliary inside-air inlet hole **118**. Here, the third adjusting door **193** may have the shape of a sliding door.

(30) In addition, during maximum heating, the second adjusting door **192** may block the second outside-air inlet **114b** and open the engine room air inlet **115** to introduce the air in the engine

room, and the third adjusting door **193** may open the auxiliary inside-air inlet hole **118** to introduce some of the inside air.

(31) In addition, the air conditioning case **110** may further include a filter **180** installed behind the first adjusting door **191** and the second adjusting door **192** in an air flow direction.

(32) In addition, the air conditioning apparatus **100** for a vehicle of the present disclosure may further include an auxiliary heating heat exchanger **170** installed behind the internal heat exchanger **120** in the air flow direction.

(33) In addition, in the air conditioning apparatus **100** for a vehicle, the blower **160** may be one component, and simultaneously blow air to the first area **A1** and the second area **A2**.

(34) In addition, an air outlet **117** through which air in the second area **A2** is discharged to the outside may be formed in the air conditioning case **110**, and the heat exchanger **140** may be installed adjacent to the air outlet **117** installed behind the blower **160** in the air flow direction.

(35) In addition, based on a dash panel **2000** partitioning the inside of the vehicle and the engine room from each other, the evaporator **130**, the heat exchanger **140** and the blower **160** may be installed in the engine room side in the air conditioning case **110**.

Advantageous Effects

(36) As set forth above, the present disclosure may provide the air conditioning apparatus for a vehicle, in which while blown air passes through a first area and a second area which are partitioned from each other by a first separation wall, air passing through the first area is cooled or heated and then supplied to the inside of a vehicle, and air passing through the second area is discharged to the outside, so that the air conditioning apparatus may be miniaturized and may easily cool or heat the inside of the vehicle.

(37) In more detail, the present disclosure may provide the air conditioning apparatus for a vehicle, in which: one blower simultaneously blows air to the first area and the second area; the first area communicates with the inside of the vehicle, and the evaporator and the internal heat exchanger are installed on the inside to cool and heat the inside; and the second area communicates with the outside, the heat exchanger is installed on the inside, and discarded air passing through the heat exchanger by the cooling and heating settings is discharged to the outside without a separate door, thereby having a small size.

(38) In addition, the present disclosure may provide the air conditioning apparatus for a vehicle, which is installed in the mounting hole of the dash panel, more easily mounted by having the evaporator, the heat exchanger and the blower installed in the engine room side in the air conditioning case and by having the internal heat exchanger disposed on the inside of the vehicle, and effectively utilizes the space in the engine room.

(39) In addition, the present disclosure may provide the air conditioning apparatus for a vehicle, in which the outside air or the inside air is selectively supplied to the first area through the first adjusting door of the air conditioning case, and the outside air is supplied to the second area through the second adjusting door.

(40) Here, the present disclosure may provide the air conditioning apparatus for a vehicle, in which the engine room air inlet is formed to be opened and closed so that air in the engine room is introduced through the second adjusting door, and during the maximum heating, the air in the engine room, which has a higher temperature than the outside, is used to enhance the evaporation performance (heat absorption amount) of the heat exchanger, thereby further enhancing the heating performance of the internal heat exchanger.

(41) In addition, the present disclosure may provide the air conditioning apparatus for a vehicle, in which the auxiliary inside-air inlet hole additionally supplying the inside air to the second area and the third adjusting door adjusting its flow are formed, and during the maximum heating, the inside warm air is used to further enhance the evaporation performance of the heat exchanger.

(42) In addition, the present disclosure may provide the air conditioning apparatus for a vehicle, in

which the auxiliary heating heat exchanger is additionally installed behind the internal heat exchanger, thereby securing the faster and more effective heating performance.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a view showing a conventional vehicle heat pump system.
- (2) FIGS. 2 through 4 are a perspective view, a cross-sectional view in direction AA' and a cross-sectional view in direction BB', each showing an air conditioning apparatus for a vehicle according to the present disclosure.
- (3) FIG. 5 is a view showing another example of the air conditioning apparatus for a vehicle according to the present disclosure.
- (4) FIG. 6 is a view showing another example of the air conditioning apparatus for a vehicle according to the present disclosure.
- (5) FIG. 7 is a view showing an example in which the air conditioning apparatus according to the present disclosure is applied to the vehicle heat pump system.
- (6) FIGS. 8, 9A and 9B are views each showing a state in which the air conditioning apparatus for a vehicle cools air according to the present disclosure.
- (7) FIGS. 10 through 12B are views each showing a state in which the air conditioning apparatus for a vehicle heats air according to the present disclosure.
- (8) FIG. 13 is a view showing a state of the heat pump system while the air conditioning apparatus performs dehumidification heating according to the present disclosure.
- (9) FIG. 14 is a cross-sectional view showing another example of the air conditioning apparatus according to the present disclosure.
- (10) FIG. 15 is a view showing an air flow in a second area of the cross-sectional view shown in FIG. 14.
- (11) FIGS. 16 and 17 are a perspective view and a cross-sectional view in direction CC' each showing another example of the air conditioning apparatus for a vehicle according to the present disclosure.
- (12) FIGS. 18 and 19 are views each showing another example and an air flow in the air conditioning apparatus for a vehicle according to the present disclosure.
- (13) FIG. 20 is a view showing another example of the air conditioning apparatus for a vehicle according to the present disclosure.
- (14) TABLE-US-00001 *Description of Reference Numerals* 1000: heat pump system L1: refrigerant circulation line, L2: coolant circulation line L3: bypass line, V: non-return valve 100: air conditioning apparatus A1: first area, A2: second area 110: air conditioning case, 110w: partition wall 110d: temp door 111: face vent, 111d: face vent door 112: defrost vent, 112d: defrost vent door 113: floor vent, 113d: floor vent door 114a: first outside-air inlet, 114b: second outside-air inlet 115: engine room air inlet 116: inside-air inlet 117: air outlet 118: auxiliary inside-air inlet hole 120: internal heat exchanger 130: evaporator 140: heat exchanger 150: second expansion member 160: blower 170: auxiliary heating heat exchanger 180: filter 191: first adjusting door, 192: second adjusting door 193: third adjusting door 200: compressor 300: first expansion member 400: water-cooled condenser, 410: gas-liquid separator 510: blocking plate 520: extension portion 530: guide portion P1: engine room air inlet passage P2: inner passage of second area 2000: dash panel, 2100: mounting hole 2200: inside-air introduction hollow portion 3000: radiator 4000: cooling fan 5000: coolant pump

BEST MODE

- (15) Hereinafter, an air conditioning apparatus **100** for a vehicle of the present disclosure having the above-described configuration is described in detail with reference to the accompanying

drawings.

(16) The air conditioning apparatus **100** for a vehicle of the present disclosure is an air conditioning apparatus **100** used for a heat pump system **1000** installed in the vehicle. The description below will again describe the heat pump system **1000** including a compressor **200**, a heat exchanger **140**, expansion members **300** and **150**, an evaporator **130** and an internal heat exchanger **120**, which are connected to one another by a refrigerant circulation line **L1**.

(17) FIGS. **2** through **4** are a perspective view, a cross-sectional view in direction **AA'** and a cross-sectional view in direction **BB'**, each showing an air conditioning apparatus for a vehicle according to the present disclosure.

(18) The air conditioning apparatus **100** for a vehicle of the present disclosure may include an air conditioning case **110**, a blower **160**, an evaporator **130**, an internal heat exchanger **120** and a heat exchanger **140**.

(19) The air conditioning case **110** may be a basic body forming the air conditioning apparatus **100** for a vehicle of the present disclosure, and may have the other components built therein, and may have a first area **A1** and a second area **A2** partitioned from each other by a partition wall **110w** in a width direction of the vehicle. Here, the first area **A1** and the second area **A2** may be partitioned from each other by the partition wall **110w** so that air is introduced thereinto and moved thereto. The air introduced into the first area **A1** may be cooled or heated to cool or heat the inside of a vehicle, and the air introduced into the second area **A2** may communicate with the outside and may be discarded to the outside.

(20) In the air conditioning case **110**, a first outside-air inlet **114a** and an inside-air inlet **116** may be formed on a portion into which the air in the first area **A1** is introduced, a temp door **110d** mixing the air passing through the evaporator **130** and the internal heat exchanger **120** based on a set temperature may be built therein, and vents **111**, **112** and **113** through which air is discharged to the inside of a vehicle are formed on a portion to which the air in the first area **A1** is discharged. Due to a dash panel **2000** partitioning the inside of the vehicle and the engine room from each other, a certain area of the air conditioning case **110** may be disposed on an engine room, and the other area may be disposed on the inside of the vehicle. A portion on which the vents **111**, **112** and **113** are formed by passing through a mounting hole **2100** of the dash panel **2000** may be positioned on the inside of the vehicle. In more detail, in the air conditioning apparatus **100** for a vehicle of the present disclosure, the evaporator **130**, the heat exchanger **140** and the blower **160** may be installed in the engine room side in the air conditioning case **110**.

(21) The first outside-air inlet **114a** is a portion which is formed by making the certain area of the air conditioning case **110** hollow and into which outside air is introduced, and the inside-air inlet **116** may be a portion which is made hollow to communicate with an inside-air introduction hollow portion **2200** of the dash panel **2000** and into which air inside the vehicle is introduced. Here, a first adjusting door **191** may adjust air flows in the first outside-air inlet **114a** and the inside-air inlet **116**.

(22) In addition, the portion to which the air in the first area **A1** is discharged may include the face vent **111**, the floor vent **113** and the defrost vent **112** whose opening degrees are respectively adjusted by doors **111**, **112d** and **113d**.

(23) Meanwhile, in the air conditioning case **110**, a second outside-air inlet **114b** may be formed on a portion into which air in the second area **A2** is introduced, and an air outlet **117** through which air is discharged to the outside may be formed on a portion to which the air in the second area **A2** is discharged. That is, all air introduced into the second area **A2** may be discharged, and the air outlet **117** needs not to have a separate door to adjust its flow.

(24) The second outside-air inlet **114b** may be a portion into which the outside air is introduced by a certain hollow area of the air conditioning case **110**, and may preferably be formed to be parallel to the first outside-air inlet **114a**. Here, a second adjusting door **192** may adjust the air flow in the second outside-air inlet **114b**.

(25) In addition, the air conditioning apparatus **100** for a vehicle of the present disclosure may further include an engine room air inlet **115** formed to be hollow. The engine room air inlet **115** may be a portion which is formed by making a certain area of the air conditioning case **110** hollow to allow air in the engine room to be introduced into the second area **A2**, and the second adjusting door **192** may adjust the flow of the outside air or the air in the engine room to be selectively introduced into the second area **A2**.

(26) In winter, when an outside temperature is low, the inside of the engine room may be heated by the operations of various electrical components and may thus maintain a higher temperature than the outside temperature. During maximum heating, the heat pump system **1000** of the present disclosure may supply the air in the engine room to the second area **A2** through the second adjusting door **192** to enhance an evaporation performance (heat absorption amount) of the heat exchanger **140**, thereby further enhancing a heating performance of the internal heat exchanger **120**.

(27) In addition, in the air conditioning case **110**, it is preferable that the portion into which the air is introduced extends to correspond to the inside-air introduction hollow portion **2200** of the dash panel **2000**, which is made hollow to introduce the inside air thereinto, and the first outside-air inlet **114a** and the second outside-air inlet **114b** are formed on an upper side of the extending portion, and the engine room air inlet **115** is formed on a lower side of the extending portion.

(28) That is, in the portion of the air conditioning case **110**, into which the air is introduced, the inside-air inlet **116** may be made hollow in a direction toward the dash panel **2000**, the first outside-air inlet **114a** and second outside-air inlet **114b** may be made hollow on the upper side, the engine room air inlet **115** may be formed on the lower side, the first outside-air inlet **114a** and inside-air inlet **116** of the first area **A1** may each have an opening degree adjusted by the first adjusting door **191**, and the second outside-air inlet **114b** and engine room air inlet **115** of the second area **A2** may each have an opening degree adjusted by the second adjusting door **192**.

(29) Meanwhile, FIG. 5 is a view showing another example of the air conditioning apparatus **100** for a vehicle according to the present disclosure; and FIG. 6 is a view showing another example of the air conditioning apparatus **100** for a vehicle according to the present disclosure. The air conditioning apparatus **100** for a vehicle of the present disclosure may allow inside air to be additionally introduced into the second area **A2** to further enhance the heating performance. The air conditioning case **110** may include an auxiliary inside-air inlet hole **118** which is formed by making a certain area hollow to allow the inside air to be introduced into the second area **A2**, and opening and closing of which is adjusted by a third adjusting door **193**.

(30) FIG. 5 shows a shape of the auxiliary inside-air inlet hole **118** which is formed by making a certain area of the partition wall **110w** hollow, and some of the air introduced into the first area **A1** through the inside-air inlet **116** may be moved to the second area **A2** through the auxiliary inside-air inlet hole **118**. When the auxiliary inside-air inlet hole **118** is formed in the partition wall **110w**, the third adjusting door **193** may preferably have the shape of a sliding door.

(31) FIG. 6 shows an example in which the auxiliary inside-air inlet hole **118** is formed by making a certain area of the wall facing the inside-air introduction hollow portion of the dash panel **2000** hollow, and a lower side of the third adjusting door **193** is pivoted as a central axis to open and close the auxiliary inside-air inlet hole **118**.

(32) During maximum heating, the second adjusting door **192** may block the second outside-air inlet **114b** and open the engine room air inlet **115** to introduce the air in the engine room into the second area **A2**, and the third adjusting door **193** may open the auxiliary inside-air inlet hole **118** to introduce some of the inside air. That is, when heating the inside of the vehicle, the air conditioning apparatus **100** for a vehicle of the present disclosure may further enhance its heating performance because the air in the engine room and inside air having a higher temperature than the outside air are supplied to the second area **A2**.

(33) The blower **160** may be installed in the air conditioning case **110** to blow air to the first area

A1 and the second area A2. In the present disclosure, the blower 160 may be one component, and operated by one motor. Through this configuration, the blower 160 may simultaneously blow air to the first area A1 and the second area A2.

(34) The evaporator 130 may be installed in the first area A1 in the air conditioning case 110 to cool air discharged to the inside of the vehicle. All air introduced into the first area A1 may pass through the evaporator 130, and when cooling is performed, a low-temperature, low-pressure, wet-saturated refrigerant may be supplied to the evaporator 130 and air may thus be cooled while passing through the evaporator 130 and discharged to the inside of the vehicle. In addition, when heating is performed, the refrigerant is not supplied to the evaporator 130, and the temperature is not changed even though air passes through the evaporator 130.

(35) The internal heat exchanger 120 may be installed behind the evaporator 130 in the first area A1 in the air conditioning case 110 to heat the air discharged to the inside of the vehicle. That is, the internal heat exchanger 120 is a component performing the heating, and the temp door 110d may adjust the cooling and the heating by adjusting the flow through which the air in the second area A2 passes through the internal heat exchanger 120.

(36) In addition, the air conditioning apparatus 100 for a vehicle of the present disclosure may further include an auxiliary heating heat exchanger 170 installed behind the internal heat exchanger 120 in an air flow direction. The auxiliary heating heat exchanger 170 may perform the heating together with the internal heat exchanger 120, and may use various forms including a positive temperature coefficient (PTC) heating member.

(37) The heat exchanger 140 may be installed in the second area A2 in the air conditioning case 110 to condense the refrigerant based on a cooling setting or to evaporate the refrigerant based on a heating setting. That is, the heat exchanger 140 does not cool or heat air for actual cooling and heating, but may condense the refrigerant supplied to the evaporator 130 based on the cooling setting or evaporate the refrigerant supplied to the internal heat exchanger 120 based on the heating setting. The description below will again describe the flow of the refrigerant based on the cooling and heating settings of the actual heat pump system 1000 and the change in the refrigerant as the refrigerant passes through each component.

(38) In addition, the heat exchanger 140 may be installed adjacent to the air outlet 117 installed behind the blower in the air flow direction. In addition, the air introduced into the second area A2 through the blower 160 may pass through the heat exchanger 140, and all of the air may then be discharged to the outside through the air outlet 117.

(39) That is, the evaporator 130 for the cooling and the internal heat exchanger 120 for the heating may be installed in the first area A1 of the air conditioning case 110, and the heat exchanger 140 condensing the refrigerant during the cooling and evaporating the refrigerant during the heating may be installed in the second area A2. Here, when wind passing through the first area A1 is cold wind for the cooling, wind passing through the second area A2 may be wind having a higher temperature than the cold wind, and when the wind passing through the first area A1 is warm wind for the heating, the wind passing through the second area A2 may be wind having a lower temperature than the warm wind.

(40) In addition, the air conditioning apparatus 100 for a vehicle of the present disclosure may further include a filter 180 installed behind the first adjusting door 191 and the second adjusting door 192 in the air flow direction, and the filter 180 may be a replaceable type.

(41) FIG. 7 is a view showing an example in which the air conditioning apparatus 100 according to the present disclosure is applied to the vehicle heat pump system 1000, and the air conditioning apparatus 100 of the present disclosure may be used for the heat pump system 1000.

(42) The heat pump system 1000 may include the compressor 200, the first expansion member 300 and the second expansion member 150 in addition to the internal heat exchanger 120, the evaporator 130, the heat exchanger 140 and the blower 160, as described above.

(43) First, the compressor 200 may be installed on the refrigerant circulation line L1 through which

the refrigerant is circulated to compress and discharge the refrigerant.

(44) The internal heat exchanger **120** may be installed in the air conditioning case **110** to exchange heat between air in the air conditioning case **110**, i.e. air conditioning wind and the refrigerant discharged from the compressor **200**, and supply the heated air conditioning wind to the inside as a result of the condensation to perform the heating.

(45) The evaporator **130** may be installed in the air conditioning case **110** to exchange heat between the air in the air conditioning case **110** and the refrigerant supplied to the compressor **200**, and evaporate the low-pressure liquid refrigerant to heat the air supplied to the inside, thereby performing the cooling.

(46) The heat exchanger **140** may be installed in the air conditioning case **110** to exchange heat between the refrigerant circulated in the refrigerant circulation line **L1** and the air. The heat exchanger **140** may condense the refrigerant during the cooling or evaporate the refrigerant during the heating.

(47) The first expansion member **300** may be installed on the refrigerant circulation line **L1** at the outlet side of the internal heat exchanger **120** to selectively expand the refrigerant discharged from the internal heat exchanger **120**, and during the heating, the first expansion member **300** may be operated to condense the refrigerant in the internal heat exchanger **120** and to evaporate the refrigerant from the heat exchanger **140**, thereby throttling the refrigerant to have a low temperature and a low pressure. In addition, during the cooling, the first expansion member **300** does not throttle the refrigerant but may bypass the refrigerant.

(48) The second expansion member **150** may be installed on the refrigerant circulation line **L1** at the inlet side of the evaporator **130** to expand the refrigerant supplied to the evaporator **130**. During the cooling, the second expansion member **150** may throttle the refrigerant condensed while passing through the heat exchanger **140** to have the low temperature and the low pressure and then supply the refrigerant to the evaporator **130**. In addition, during the heating, the second expansion member **150** does not throttle the refrigerant but may bypass the refrigerant.

(49) In addition, the vehicle heat pump system **1000** may further include a water-cooled condenser **400** installed on the refrigerant circulation line **L1** at the inlet side of the first expansion member **300** outside the air conditioning case **110** to exchange heat with a coolant cooling a battery. During the cooling, the water-cooled condenser **400** is a portion in which the coolant flows along a coolant circulation line **L2** to perform the heat exchange, and may condense the refrigerant supplied to the evaporator **130** together with the internal heat exchanger **120** and the heat exchanger **140** based on the cooling setting. A radiator **3000** cooling the coolant and a coolant pump **5000** circulating the coolant may be installed on the coolant circulation line **L2**, and a cooling fan **4000** blowing air to the radiator **3000** may be installed adjacent to the radiator **3000**. The water-cooled condenser **400** may include a gas-liquid separator **410**. In this case, the refrigerant may be condensed by the water-cooled condenser **400** and the liquid refrigerant separated by the gas-liquid separator **410** may be overcooled by the heat exchanger **140**. Therefore, it is possible to enhance the cooling performance of the evaporator **130** by enhancing its condensing performance.

(50) In addition, the vehicle heat pump system **1000** may have a bypass line **L3** formed on the refrigerant circulation line **L1** to bypass the second expansion member **150** and the evaporator **130**, and a non-return valve **V** may be additionally installed on the bypass line **L3** to prevent reverse flow of the refrigerant. During the heating, on the bypass line **L3**, the refrigerant does not pass through the second expansion member **150** or the evaporator **130**, but may pass through the heat exchanger **140** to be supplied to the compressor **200**. That is, it is possible to omit unnecessary components (second expansion member **150** and evaporator **130**) during the heating to prevent unnecessary flow rate decrease and pressure drop.

(51) FIGS. **8**, **9A**, and **9B** are views each showing a state in which the air conditioning apparatus for a vehicle cools air according to the present disclosure.

(52) First, during the maximum heating, the refrigerant compressed by the compressor **200** may

pass through the internal heat exchanger **120** without heat exchange (because the temp door **110d** blocks the flow of air passing through the internal heat exchanger **120** in the first area **A1**), and the high-pressure refrigerant delivered from the compressor **200** may be condensed while passing through the water-cooled heat exchanger **140** and the heat exchanger **140**. Here, the first expansion member **300** may bypass the refrigerant. The condensed refrigerant may be throttled in the second expansion member **150** and supplied to the evaporator **130**, and heat may be exchanged between the low-pressure liquid refrigerant throttled in the evaporator **130** and the air passing through the first area **A1**. Therefore, air discharged to the inside may be cooled by heat absorption by the evaporative latent heat of the refrigerant.

(53) Here, in order to increase cooling efficiency, the first adjusting door **191** may close the first outside-air inlet **114a** and open the inside-air inlet **116**, and the inside air may thus be cooled by exchanging heat with the evaporator **130** while being introduced into and moved to the first area **A1**. The second adjusting door **192** may open the second outside-air inlet **114b**, and the outside air may thus be discharged by exchanging heat with the heat exchanger **140** while being introduced into and moved to the second area **A2**.

(54) Meanwhile, dehumidification cooling may be performed in the same state as the cooling state shown in FIGS. **8**, **9A**, and **9B**, but may be performed by adjusting the temperature of air supplied to the inside of the vehicle by adjusting the temp door **110d**.

(55) FIGS. **10** through **12B** are views each showing a state in which the air conditioning apparatus for a vehicle heats air according to the present disclosure.

(56) First, during the maximum heating, the refrigerant compressed by the compressor **200** may exchange heat with the air passing through the first area **A1** while passing through the internal heat exchanger **120**, and may thus heat the air discharged to the inside due to the heat dissipation by its condensation. Meanwhile, the air introduced into the first area **A1** may pass through the evaporator **130**, the temp door may be operated to allow all the air to pass through the internal heat exchanger **120**. Here, the air may pass through the evaporator **130** without heat exchange with the evaporator **130** in a state in which no refrigerant is supplied to the evaporator **130**. The coolant does not flow along the coolant circulation line **L2** in the water-cooled condenser **400** either, and the refrigerant may be moved without change. The refrigerant may be throttled in the first expansion member **300**, supplied to the heat exchanger **140**, and then evaporated. The refrigerant evaporated while passing through the heat exchanger **140** may be supplied to the compressor **200** through the bypass line **L3** without passing through the second expansion member **150** or the evaporator **130**.

(57) Here, the first adjusting door **191** may selectively open the first outside-air inlet **114a** or the inside-air inlet **116** to blow the air into the first area **A1**, and the second adjusting door **192** may close the second outside-air inlet **114b** and open the engine room air inlet **115** to allow the air in the engine room to be introduced into the second area **A2**. In addition, the third adjusting door **193** may open the auxiliary inside-air inlet hole **118** to allow the inside air to be introduced thereinto. Here, FIGS. **11A**, **11B**, **12A**, and **12B** show cases in which different third adjusting doors **193** are formed, in which all the third adjusting doors **193** open the auxiliary inside air inlet hole **118**.

(58) Meanwhile, FIG. **13** is a view showing a state of the heat pump system **1000** while the air conditioning apparatus **100** performs dehumidification heating according to the present disclosure, which shows an example in which the air passing through the heat exchanger **140** does not pass through the bypass line **L3** but may pass through the second expansion member **150** and the evaporator **130** to additionally absorb heat, thereby adjusting dehumidification and temperature of the heat pump system.

(59) In summary, the air conditioning apparatus **100** for a vehicle of the present disclosure may be the air conditioning apparatus **100** used for the heat pump system, that is, an air conditioning system which may perform the cooling and the heating using one refrigerant line. The heating may use the high-temperature refrigerant compressed by the compressor **200**, and the cooling may use the evaporation of the refrigerant. Here, a chronic problem of the heat pump system, i.e.

deteriorated condensing performance during the cooling may be solved by using the heat exchanger **140** condensing the refrigerant together with the internal heat exchanger **120**. The heat exchanger **140** may be mounted in the air conditioning case **110**, and condense the refrigerant by receiving the wind from the blower **160** generating the air conditioning wind. That is, the heat exchanger **140** does not have a separate fan or blower blowing the air, and may receive the wind from the blower **160** generating the air conditioning wind, thereby reducing the overall size of the vehicle.

(60) In addition, the air conditioning apparatus **100** for a vehicle of the present disclosure may be formed in a structure in which the air discharged from the second area **B2** is not introduced back into the second area **B2**. During the heating, air discharged to the air outlet **117** by passing through the heat exchanger **140** of the second area **B2** may exchange heat with the heat exchanger **140** to have a lower temperature than the air in the engine room. Accordingly, if the air discharged from the second area **B2** is again supplied to the second area **B2** through the engine room air inlet **115**, it is difficult to fully enhance the heating performance by introducing the high temperature air of the engine room. In order to solve this problem, the air conditioning apparatus **100** for a vehicle of the present disclosure may have a structure in which air discharged from the second area **B2** is not introduced back into the second area **B2**, and may have various forms.

(61) First, FIGS. **14** and **15** each show an example in which a blocking plate **510** blocking the air flow in a height direction is formed by connecting between the air outlet **117** of the air conditioning case **110** and the dash panel **2000**. That is, the blocking plate **510** may block the air discharged through the air outlet **117** from being moved to an engine room air inlet passage **P1**.

(62) FIGS. **16** and **17** are a perspective view and a cross-sectional view in direction **CC'** each showing another example of the air conditioning apparatus **100** for a vehicle according to the present disclosure; FIGS. **18** and **19** are views each showing another example and an air flow in the air conditioning apparatus for a vehicle according to the present disclosure; and FIG. **20** is a view showing another example of the air conditioning apparatus **100** for a vehicle according to the present disclosure.

(63) First, FIGS. **16** and **17** each show that an extension portion **520** is additionally formed by extending a periphery of the blocking plate **510** in an upward direction to surround the air outlet **117**.

(64) In addition, FIGS. **18** and **19** show an example in which a guide portion **530** is formed on a portion of the air outlet **117**, which is adjacent to the dash panel **2000**, to guide the air discharged through the air outlet **117** in a direction away from the dash panel **2000**.

(65) In addition, FIG. **20** shows an example in which an inner passage **P2** of the second area **100B**, which is a space in the second area **100B**, and the engine room air inlet passage **P1** are formed to be parallel to each other. In more detail, in the air conditioning case **110**, the second outside-air inlet **114b** disposed above the second area **A2**, and the air outlet **117** disposed therebelow, may be respectively disposed at the upper and lower sides in the height direction. That is, the second outside-air inlet **114b** may be disposed in an area upwardly extending from an area where the air outlet **117** is formed.

(66) The air conditioning apparatus **100** for a vehicle of the present disclosure, which has such a configuration, may allow the air in the engine room, having a high temperature, to be supplied to the second area **A2** through the engine room air inlet **115**, thereby sufficiently securing its heating performance.

(67) The air conditioning apparatus **100** for a vehicle of the present disclosure may be formed in the shapes shown in FIGS. **2** through **20**, and may be formed in more various shapes as long as the apparatus has a structure in which the air discharged from the second area **A2** is prevented from being introduced back to the engine room air inlet **115** of the second area **A2**.

(68) The present disclosure is not limited to the above-mentioned embodiments, and may be variously applied. In addition, the present disclosure may be variously modified by those skilled in

the art to which the present disclosure pertains without departing from the gist of the present disclosure claimed in the claims.

Claims

1. An air conditioning apparatus for a vehicle, comprising: an air conditioning case including a first area which communicates with the inside of the vehicle and in which inside-air conditioning wind flows and a second area which communicates with the outside of the vehicle; a blower which blows the wind to the first area and the second area; and an auxiliary inside-air inlet hole formed by hollowing out a region to allow internal air to enter the second region, and abutting a third adjustable door, the opening and closing of which is controlled by said third adjustable door, and the auxiliary inside-air inlet hole is adjacent to an inside-air inlet and is formed by hollowing out a portion of a partition wall, which is a partition wall separating a first area and a second area, and moving a portion of the air entering the first area through said inside-air inlet to the second area via said auxiliary inside-air inlet, wherein the air entering the second area is not cooled and heated and is discharged to the outside of the vehicle.
2. The air conditioning apparatus for a vehicle of claim 1, wherein the blower is a blower operated by one motor, and the wind is blown to the first area and the second area by the blower.
3. The air conditioning apparatus for a vehicle of claim 1, comprising an evaporator for a cooling and an internal heat exchanger for a heating in the first area of the air conditioning case.
4. The air conditioning apparatus for a vehicle of claim 3, comprising a heat exchanger installed in the second area of the air conditioning case to condense a refrigerant during the cooling and to evaporate the refrigerant during the heating.
5. The air conditioning apparatus for a vehicle of claim 4, wherein when wind passing through the first area is cold wind for the cooling, wind passing through the second area is wind having a higher temperature than the cold wind.
6. The air conditioning apparatus for a vehicle of claim 5, wherein when the wind passing through the first area is warm wind for the heating, the wind passing through the second area is wind having a lower temperature than the warm wind.
7. The air conditioning apparatus for a vehicle of claim 6, wherein outside air or air in an engine room is selectively introduced into the second area.
8. The air conditioning apparatus for a vehicle of claim 7, wherein the air conditioning case includes: a first outside-air inlet into which outside air is introduced and an inside-air inlet into which inside air is introduced, each of which is formed by making a certain area of the air conditioning case hollow; a first adjusting door adjusting openings and closings of the first outside-air inlet and the inside-air inlet; a second outside-air inlet which is formed in the second area of the air conditioning case and into which the outside air is introduced; and a second adjusting door adjusting opening and closing of the second outside-air inlet.
9. The air conditioning apparatus for a vehicle of claim 8, wherein the air conditioning case includes an engine room air inlet which is formed by making a certain area hollow to allow the air in the engine room to be introduced into the second area, and opening and closing of which is adjusted by the second adjusting door.
10. The air conditioning apparatus for a vehicle of claim 9, wherein based on a dash panel partitioning the inside of the vehicle and the engine room from each other, the evaporator, the heat exchanger and the blower are installed in the engine room side in the air conditioning case.
11. The air conditioning apparatus for a vehicle of claim 10, wherein in the air conditioning case, a portion into which the air is introduced extends to correspond to an inside-air introduction hollow portion of the dash panel, which is made hollow to introduce the inside air thereinto, and the first outside-air inlet and the second outside-air inlet are formed on an upper side of the extending portion, and the engine room air inlet is formed on a lower side of the extending portion.

12. The air conditioning apparatus for a vehicle of claim 10, wherein an air outlet through which air in the second area is discharged to the outside is formed in the air conditioning case, and the heat exchanger is installed adjacent to the air outlet installed behind the blower in the air flow direction.
 13. The air conditioning apparatus for a vehicle of claim 12, wherein the air discharged from the second area is not introduced back into the second area.
 14. The air conditioning apparatus for a vehicle of claim 13, wherein a blocking plate blocking an air flow in a height direction is formed by connecting between the air outlet of the air conditioning case and the dash panel.
 15. The air conditioning apparatus for a vehicle of claim 14, wherein an extension portion is additionally formed by extending a periphery of the blocking plate in an upward direction.
 16. The air conditioning apparatus for a vehicle of claim 13, wherein a guide portion is formed on a portion of the air outlet, which is adjacent to the dash panel, to guide the air discharged through the air outlet in a direction away from the dash panel.
 17. The air conditioning apparatus for a vehicle of claim 13, wherein the second outside-air inlet and the air outlet are respectively disposed at the upper and lower sides in the height direction, thereby forming an engine room air inlet passage, which is a space through which the air in the engine room between the air conditioning case and the dash panel is moved to the engine room air inlet, and an inner passage of the second area, which is a space in the second area, to be parallel to each other.
 18. The air conditioning apparatus for a vehicle of claim 8, wherein the air conditioning case further includes a filter installed behind the first adjusting door and the second adjusting door in an air flow direction.
 19. The air conditioning apparatus for a vehicle of claim 4, further comprising an auxiliary heating heat exchanger installed behind the internal heat exchanger in the air flow direction.
 20. The air conditioning apparatus for a vehicle of claim 1, wherein the third adjusting door has the shape of a sliding door.
 21. The air conditioning apparatus for a vehicle of claim 1, wherein during maximum heating, the second adjusting door blocks the second outside-air inlet and opens the engine room air inlet to introduce the air in the engine room, and the third adjusting door opens the auxiliary inside-air inlet hole to introduce some of the inside air.
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